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Meta-Analysis

Comprehensive Comparisons of Different Treatments for Active Graves Orbitopathy: A Systematic Review and Bayesian Model–Based Network Meta-Analysis

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Abstract

Context: Graves orbitopathy is a specialized immunoinflammatory disorder related to abnormal thyroid function. Due to the complexity of the disease and its propensity to reoccur, targeted treatment is essential to improve the symptoms.

Objective: This study aims to assess the efficacy and safety of various treatments for active thyroid eye disease.

Methods: We conducted a comprehensive search for randomized controlled trials (RCTs), and ongoing RCTs registered on Controlled Trials, targeting treatments for thyroid eye disease until November 20, 2024. Employing a Bayesian framework, this network meta-analysis calculated risk ratios (RRs) or mean differences (MDs) with 95% CIs to size the effects for the predetermined outcomes. The study is registered with PROSPERO (CRD42024548030).

Outcome: The primary outcomes evaluated were overall response rate, clinical activity score (CAS), proptosis, diplopia, and adverse events.

Results: For the overall response rate, teprotumumab (RR 5.5, 95% CI 2.3 to 16), mycophenolate combined intravenous glucocorticosteroids (IVGCs) demonstrated effectiveness over no treatment, ranked from most to least effective. Notably, teprotumumab showed the highest efficacy in reducing CAS (MD −1.57, 95% CI −3.81 to 0.68) and proptosis (MD −2.29, 95% CI −2.73 to −1.86). For diplopia improvement, teprotumumab and IVGCs were effective compared with no treatment.

Conclusion: Teprotumumab emerges as potentially the most effective treatment for reducing inflammation and increasing overall response rates when compared with no treatment; oral mycophenolate combined with IVGCs appears to be the best for improving proptosis. While some treatments raise safety concerns due to reported adverse events, oral methotrexate combined with IVGCs appear to offer a favorable balance between efficacy and safety among the evaluated treatments.

Key Words: Graves orbitopathy, nonsurgical treatments, comprehensive efficacy, monoclonal antibody

Abbreviations: CAS, Clinical Activity Score; CT, computed tomography; GO, Grave's orbitopathy; IVGC, intravenous glucocorticosteroid; MD, mean difference; MM, mycophenolate; OGC, oral glucocorticosteroid; RCT, randomized controlled trial; RR, risk ratio; SUCRA, surface under the cumulative ranking curve.

Thyroid-associated ophthalmopathy, also known as Graves orbitopathy (GO), is a specialized immunoinflammatory disorder affecting the orbital and periorbital tissues. It is predominantly linked with autoimmune thyroid disease and stands as the most prevalent extrathyroidal manifestation of Graves disease (1). The clinical presentation of GO includes proptosis (eye bulging), eyelid swelling, restricted eye movement, and visual disturbances (2). In its severe forms, people with GO can escalate to optic neuropathy or complete vision loss, significantly diminishing the quality of life.

The pathophysiology of GO is complex and not entirely understood, but it is widely recognized to involve autoimmune-induced inflammation of the orbital tissues (3), increased activity and proliferation of thyroid-associated fibroblasts (2, 4), and enhanced lipogenesis (5). Consequently, the primary objectives in managing GO are to suppress autoimmune reactions, reduce orbital inflammation, and minimize tissue remodeling in the orbit.

Currently, the therapeutic arsenal for active, moderate to severe GO includes glucocorticoids (6), immunosuppressants, and radiotherapy. The European Group on Graves' Orbitopathy (EUGOGO) recommends a standard regimen of glucocorticosteroids (7), typically administered at a total dose of 4.5 g, as the first-line treatment. Although this approach is

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widely accepted and can achieve response rates between 60% and 80% (8), prolonged glucocorticoid use is associated with several adverse effects, such as osteoporosis and Cushing syndrome. Moreover, a subset of patients may not respond adequately to glucocorticoids, experience relapses, or develop intolerance to the medication due to its side effects.

Notably, teprotumumab, a human monoclonal antibody targeting the insulin-like growth factor-1 receptor, has been identified as a potent inhibitor of the pathological pathways in GO. By blocking insulin-like growth factor-1 receptor signaling (9), teprotumumab effectively prevents the inflammatory infiltration, fibrosis, and remodeling of orbital tissues, marking a significant breakthrough in the treatment landscape (10, 11).

Despite the availability of several traditional pairwise meta-analyses providing substantial evidence, these studies often limit their comparisons to 2 interventions and suffer from small sample sizes and a limited number of randomized controlled trials (RCTs). To overcome these limitations and offer an individualized treatment recommendations for active patients with GO, this study employs a network meta-analysis (12). Network meta-analysis includes direct evidence for pairwise comparisons and indirect evidence for closed-loop comparisons of network maps. Pooling of direct and indirect evidence from randomized trials, called mixed treatment comparisons. Mixed treatment comparison generates estimates of the relative impacts of each treatment compared with every other treatment in a network and allows comprehensive judgments on the rank of the measured outcomes (13). This method allows for both direct and indirect comparisons among treatments, combining results to enhance statistical power and provide a more comprehensive evaluation of treatment effects (14). By delivering up to date insights on the various treatments for patients with GO, this analysis aims to inform policy development and guide clinical practice in selecting the most appropriate therapeutic strategies (15).

Materials and Methods

Search Strategy

We conducted an exhaustive search across multiple databases, including PubMed, Embase, the Cochrane Library, and WanFang, up to November 20, 2024. The search strategy was precisely formulated to capture all relevant studies by incorporating a wide range of keywords related to GO, including "Graves Ophthalmopathy" or "thyroid-associated ophthalmopathy" or "thyroid eye disease" or "Graves' ophthalmopathy" or "Graves' orbitopathy," "glucocorticoids" or "methylprednisolone" or "orbital radiotherapy" or "mycophenolic acid" or "azathioprine" or "rituximab" or "teprotumumab" or "methotrexate" or "triamcinolone acetonide" or "tocilizumab" or "tripterygium glycosides." Details of the specific search strategy and search results are given elsewhere (Appendix S1 (16)). In addition, we searched for pairwise meta-analyses to discover any additional papers that were not included in the primary search.

Study Selection and Data Extraction

The protocol for this systematic review and network meta-analysis was registered with PROSPERO (CRD42024548030). This study followed the PRISMA Preferred Reporting Items for Systematic Reviews and Meta-analyses 2020 (17) and the network Meta-analysis Extended Statement (PRISMA-NMA) (18). The screening procedure consisted of 3 steps. Initially, 3 reviewers (X.Y.L., L.R.K., and Z.J.Y.) reviewed the titles and abstracts of clinical studies that included people diagnosed with GO as an initial screening criteria. Secondly, the papers were screened based on certain criteria for eligibility (Table 1). Thirdly, the articles that met the established criteria were subjected to further scrutiny in the browse full text. Two independent reviewers (X.Y.L. and L.R.K.) independently assessed the included studies and extracted data using a standardized extraction form. A list of data extracted from the included randomized clinical trials is shown elsewhere (Appendix S3 (16)). For missing or not directly available data, we contacted the authors of the trial reports.

Table 1. Screening criteria for clinical studies used in the paper

Screening criteria	Included	Excluded
Research type	Randomized controlled clinical studies	1. Own before and after controlled study 2. Retrospective study
Population	Active GO at study baseline	1. Chronic inactive GO 2. Received treatment for GO in the 3 months prior to study entry
Interventions	IVGCs, OGCs, tretinoin orbital injections, OR, antiproliferative drugs (mycophenolate mofetil, azathioprine, methotrexate), biologics (rituximab, teplumab, tolizumab), traditional Chinese medicine (tripterygium glycosides), combination therapy, and NT	Surgical and supporting treatment etc.
Outcomes	overall response rate, CAS, proptosis, diplopia and adverse events.	NA

Abbreviations: CAS, Clinical Activity Score; IVGC, intravenous glucocorticosteroid; GO, Graves orbitopathy; NT, no treatment; OGC, oral glucocorticosteroid; OR, orbital radiotherapy.

Outcomes of Interest

The primary outcomes of interest were meticulously chosen to reflect both the efficacy and safety of treatments:

Overall response rate

The overall response rate was defined by a composite index of improvements, including (1) improvement in clinical activity score (CAS) by at least 2 points or disease inactivation (CAS ≤ 3); (2) reduction in palpebral aperture by at least 2 mm; (3) reduction in exophthalmos by at least 2 mm; (4) improvement of soft tissue involvement by at least 1 level; (5) the GRADE for diplopia was improved by at least 1 level; (6) improvement of >8 degrees in eye muscle motility in any duction. Overall response rate was determined by meeting at least 2 of these criteria without any deterioration in ophthalmic index parameter in both eyes simultaneously.

Clinical activity score

CAS assessed the absolute change in CAS from baseline to the end of follow-up, providing an objective measure of disease activity.

Proptosis

Proptosis was assessed using computed tomography (CT) imaging to measure the absolute change in ocular protrusion from baseline to the study's conclusion.

Diplopia

Diplopia was evaluated based on improvements in the Gorman diplopia score, with responders showing at least a 1-grade improvement.

Adverse Events

The frequency and severity of adverse events and significant adverse reactions were recorded to assess treatment safety.

Quality Assessment of Evidence

The quality of included studies was assessed using the Cochrane Risk of Bias. This assessment encompassed random sequence generation, allocation method concealment, blinding of participants and researchers, completeness of outcome data, selective reporting of results, and other biases (19). Each trial was assessed whether it had a low, high, or unclear risk of bias for each area. Two reviewers completed the risk of bias assessment independently, and any disagreements were resolved by consensus. We used funnel plots to detect publication bias. To assess the presence of small study effects, we used Egger's test for continuous variables and AS-Thompson's test for dichotomous variables (20).

We also used the GRADE evidence framework to evaluate the quality of evidence, dividing each end indicator that can form a closed circle into 4 quality levels of height, moderate, low or very low. The quality of the evidence for each result is based on 5 areas: bias risk, inconsistency, indirectness, inaccuracy and public bias (21).

Data and Statistical Analysis

For outcomes that include dichotomous variables, we conducted meta-analyses of risk ratios (RRs) and odds ratios (ORs) as ORs and RRs are almost equal at lower prevalence rates (less than 10%). In this paper, RRs is comparable to ORs due to the low prevalence of the outcome. Furthermore, we found that using ORs and RRs had similar values, but the CI of ORs were larger, so we reported RRs with 95% CIs as the effect sizes. For continuous variables, we used mean differences (MDs) with 95% CIs as the measure of effect sizes for these outcomes. We used the R 4.3.3 software "gemtc" package to conduct a network meta-analysis using Bayesian models. The study utilized consistency, inconsistency and random-effects models to gather the results of relative risks and MDs to compare different treatment measures to no treatment.

Consistency testing includes global consistency and local consistency. If global consistency holds, it suggests that the results of the entire network are reliable. Global consistency is usually assessed by Markov Chain Monte Carlo method (22), which detects consistency by modeling the posterior distribution of a Bayesian model. It constructs a Markov chain with a sequence of samples that are consistent with the posterior distribution throughout multiple iterations of sampling. Finally, the convergence of the samples or the chain is checked to ensure the sampling outcomes roughly match the true posterior distribution (23). Meanwhile, we used the deviance information criterion for global consistency judgment, which is a compromise metric between fitness and complexity of Bayesian models, combining deviance and model complexity.

Local consistency between direct and indirect estimates within each closed-loop evidence was assessed using node-splitting (13). Local node consistency usually implies agreement between direct and indirect evidence in the vicinity of a particular node of network. Heterogeneity was evaluated using the I^2 statistic, where a value of 0% indicates no heterogeneity was observed, and larger values indicate increased heterogeneity (24). Mesh mapping was performed using Stata MP 18.0, and ranked plots were used surface under the cumulative ranking curve (SUCRAs) to determine the ranked probability of the best treatment .

Furthermore, to evaluate whether the initial features may impact the outcomes of the research, meta-regression analyses were conducted using the "gemtc" package to identify factors that could lead to heterogeneity. These factors included the sample size for each trial, thyroid function, follow-up time and severity of GO (25). Finally, we conducted a sensitivity analyses by excluding certain studies based on compositive data results.

Results

Literature Selection and Study Characteristics

We conducted a search of 2914 citations, then examined the titles and abstracts of 967 publications and assessed 86 full-text articles. Upon further review, 70 studies were excluded for various reasons: 34 were non-RCTs, 14 involved only a single experimental group, 10 lacked reported outcomes or the disease activity of interest, 6 included participants with chronic inactive GO, 2 have been retracted, 1 involved different populations, and 4 were lower quality (Appendix S2 (16)). Ultimately, 19 RCTs involving 1223 participants met our inclusion criteria. These studies encompassed 12 different treatment approaches, including 3 methods of administering glucocorticoids: intravenous glucocorticosteroids (IVGCs), oral glucocorticosteroids (OGCs), and retro-ocular glucocorticosteroids (triamcinolone acetonide). The sample sizes varied from 15 to 149 participants, predominantly Caucasian, with an age range of 33-60 years, a majority being female, and 38.5% (471 participants) were smokers at baseline. The duration from participant enrollment to outcome evaluation ranged between 12 and 52 weeks. Most studies recruited patients with normal thyroid function at trial commencement. The characteristics of the studies are summarized in Table 2.

Overall Response Rate

Our network meta-analysis included 13 trials with a total of 752 participants. The results illustrated in Figs. 1A and 2 show that teprotumumab significantly improved the overall response rate in adults with GO compared with placebo or sham radiotherapy, with a RR of 5.49 (95% CI 2.31 to 16.59), SUCRA 84%. Other effective treatment included mycophenolate (MMF) combined with IVGCs (MMF + IVGCs) (RR 4.91, 95% CI 1.49 to 34.05, SUCRA 83.5%). Specific SUCRA rankings can be found elsewhere (Appendix S8 (16)). Indirect comparison revealed teprotumumab's superior effectiveness compared with OR (orbital radiotherapy) (RR 2.23, 95% CI 1.01 to 14.91); MMF + IVGCs and IVGCs were both superior triamcinolone acetonide and OGCs. Oral methotrexate combined with IVGCs (oMTX + IVGCs) was superior periocular methotrexate (piMTX). The GRADE assessment indicated

Table 2. Characteristics of studies included in the network meta-analysis

Study	Randomized treatments (participants, n)	Age (mean ± SD, years)	Female (n)	Smokers (n)	Euthyroidism (n)	Duration of GO (months)	Duration of thyroid disease (months)	Ocular characteristics	Follow-up (weeks)	Outcome(s)
(26) Swaify 2023	Periocular MTX (n = 18) TG (n = 18)	40.2 ± 10	9	7	5	3.4	10.22	Moderate-severe	30	CAS, proptosis, overall response rate, diplopia
(27) Li 2023	MMF + IVGCs (n = 30) IVGCs (n = 30)	44.13 ± 11.69 41.47 ± 9.83	8 12	NA	No	11 11.67	NA	Moderate-severe, asymmetric GO	24	CAS, proptosis, overall response rate
(28) Shen 2022	Oral MTX + IVGCs (n = 30) IVGCs (n = 30)	46.5 46.5	20 21	10 13	23 23	6.5 6	NA	Moderate-severe	12	CAS, proptosis, overall response rate, diplopia
(29) Douglas 2020	Teprotumumab (n = 41) Placebo (n = 42)	51.6 ± 12.6 48.9 ± 13.0	29 31	9 8	No	6.2 6.4	42 26.4	Moderate-severe	24	proptosis, overall response rate, diplopia
(30) Kahaly 2018	MMF + IVGCs (n = 76) IVGCs (n = 73)	52.1 50.6	61 64	44 41	No	9 8.5	13.5 15	Moderate-severe	12	CAS, overall response rate
(11) Smith 2017	Teprotumumab (n = 42) Placebo (n = 45)	51.6 ± 10.6 54.2 ± 13.0	28 36	32 26	20 13	4.7 5.2	10.7 10.8	Moderate-severe	24	CAS, proptosis, overall response rate, diplopia
(31) Li 2017	OR (n = 72) OR + IVGCs (n = 70)	49.1 ± 9.3 50.3 ± 10.0	48 47	14 12	72 70	NA	NA	Active	24	CAS, proptosis,
(32) Stan 2015	Rituximab (n = 13) Placebo (n = 12)	57.6 ± 12.7 61.8 ± 11.0	8 8	2 2	No	12.4 10	NA	Moderate-severe	12	CAS, proptosis, overall response rate
(33) Salvi 2015	Rituximab (n = 15) IVGCs (n = 16)	51.9 ± 13.1 50.4 ± 11.4	14 12	10 9	No	4.5 4.6	NA	Moderate-severe	24	CAS, overall response rate
(34) Roy 2015	IVGCs (n = 31) OGC (n = 31)	37.61 ± 6.31 36.93 ± 9.44	22 16	7 13	No	15.54 12.32	NA	Moderate-severe	52	CAS, proptosis, overall response rate, diplopia
(35) Alkawas 2010	OGC (n = 12) TA (n = 12)	40 40	7 9	NA	NA	NA	NA	Active	24	CAS, proptosis, overall response rate, diplopia
(36) Geest 2008	IVGCs (n = 6) OGC (n = 9)	50.7 ± 11.5 44.7 ± 15.4	6 6	4 4	No	13 6	36 38	Moderate-severe	48	CAS, proptosis, overall response rate, diplopia
(37) Aktaran 2007	IVGCs (n = 25) OGC (n = 27)	44.3 ± 11 41.3 ± 12	14 14	10 9	No	3 3	NA	Moderate-severe	12	CAS, proptosis, overall response rate, diplopia
(38) Kahaly 2005	IVGCs (n = 35) OGC (n = 35)	52 48	25 24	21 21	No	4 3	5 4	Moderate-severe	12	CAS, proptosis, overall response rate, diplopia
(39) Ng 2005	OR + IVGCs (n = 8) IVGCs (n = 8)	64.1 ± 9.7 48.3 ± 16.0	5 1	2 1	No	2.5 3	12 10	Moderate-severe	52	proptosis
(40) Prummel 2004	OR (n = 44) Sham OR (n = 44)	45.2 ± 12.0 45.1 ± 12.8	33 37	25 22	No	17 15	21 26	Mild (proptosis value ≤24 mm)	48	CAS diplopia
(41) Marcocci 2001	OR + IVGCs (n = 41) OR + OGCs (n = 41)	50 ± 10 48 ± 10	35 33	29 25	41 41	35 34	39 43	Moderate-severe	48	CAS, proptosis, overall response rate, diplopia
(42) Gorman 2001	OR (n = 42) Sham OR (n = 42)	48	NA	NA	No	15.6	33.6	Moderate	24	proptosis
(43) Mourits 2000	OR (n = 30) Sham OR (n = 30)	48.7 ± 11.8 49.0 ± 14.2	23 28	21 18	No	13 14.5	NA	Moderate-severe	24	overall response rate, diplopia

Abbreviations: CAS, Clinical Activity Score; GO, Graves orbitopathy; IVGC, intravenous glucocorticoid; MMF, mycophenolate mofetil; oMTX, oral methotrexate; OR, orbital radiotherapy; OGC, oral glucocorticoid; piMTX, periocular methotrexate; TA, triamcinolone acetonide; TG, tripterygium glycosides.

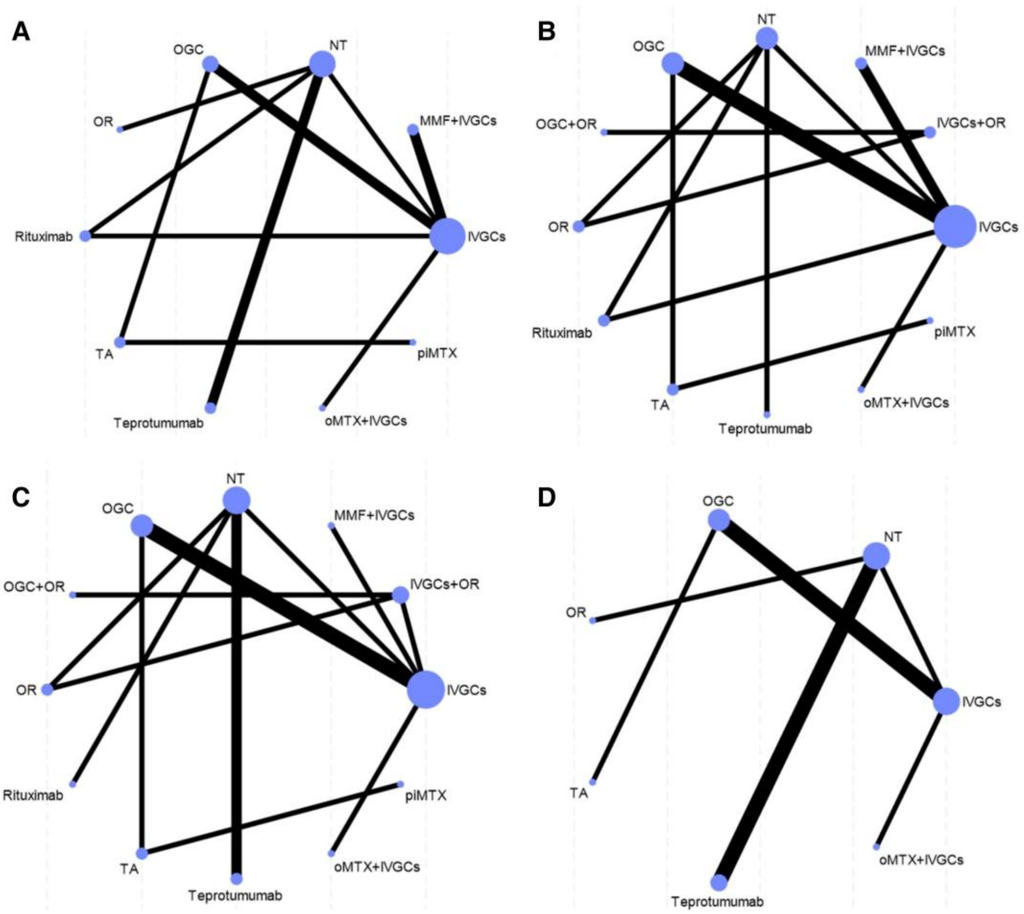

Figure 1. Network map of outcomes. Nodes: Circles representing each intervention included in the network meta-analysis; the size of the circle is proportional to the number of patients assigned to that treatment. Links: Lines connecting 2 nodes; the thickness of lines between the interventions relates to the number of studies for that comparison. A link between 2 nodes indicates that there is direct evidence for the comparison.

Abbreviations: IVGCs, intravenous glucocorticoids; OR, orbital radiotherapy; OGC, oral glucocorticoid; MMF, mycophenolate mofetil; TA, triamcinolone acetonide; piMTX, periocular methotrexate; oMTX, oral methotrexate; NT, no treatment. (A) Network map of overall response rate. (B) Network map of CAS. (C) Network map of proptosis. (D) Network map of diplopia.

that the evidence for the overall response rate was predominantly moderate to high (Appendix S9, Table S9-1 (16)).

Clinical Active Score

The Clinical Activity Score (CAS) utilizes a binary scoring system with 7 elements, each contributing 1 point to the score, a CAS of 3 or higher is indicative of active GO. Network map showed in Fig. 1B. In descending order of effectiveness, teprotumumab (SUCRA 80.7%), rituximab, OR + IVGCs, and MMF + IVGCs exerted an acceptable effect, but has not yet demonstrated a statistically significant difference. Following teprotumumab, rituximab (SUCRA 74.2%) and IVGCs + OR (SUCRA 73.2%) were ranked. As illustrated in Fig. 3, teprotumumab exhibited superior efficacy compared to OR (MD −1.93, 95% Cl −2.5 to −1.36), MMF + IVGCs and IVGCs were both effective than OR and OGC. The GRADE assessment revealed that the quality of evidence for CAS ranged from low to high, as detailed elsewhere (Appendix S9, Table S9-2 (16)).

Proptosis

Proptosis, a hallmark symptom of GO, is commonly evaluated in patients using a Hertel exophthalmometer or CT imaging to measure ocular proptosis or assess the extent and degree of eye protrusion. A network map is shown in Fig. 1C. As illustrated in Fig. 4, MMF + IVGCs (MD −2.42, 95% CI −4.83 to −0.11, SUCRA of 93%) and teprotumumab (MD 2.29 95% CI −2.73 to −1.86, SUCRA 90.8%) significantly reduced proptosis in adults with GO compared with no treatments. Further analysis revealed that MMF + IVGCs was more effective than IVGCs, OR, and triamcinolone acetonide; teprotumumab outperformed rituximab, OR, and OGCs. The GRADE assessment indicated that the overall quality of evidence for proptosis was moderate to high, as detailed in elsewhere (Appendix S9, Table S9-3 (16)).

Diplopia

Diplopia, characterized by symptoms such as double vision, blurred vision, or visual distortions, significantly affects visual function in many patients with thyroid eye disease. Significant differences were observed with teprotumumab (RR 6.53, 95% CI 3.39 to 12.55) and OR (RR 4.25 95% CI 1.27 to 18.48) compared with placebo or sham irradiation. The efficacy comparisons are illustrated in Fig. 5.

Adverse Events

In the network meta-analysis, 13 studies comparing 8 medications against placebo were included to assess adverse events. Radiation complications were scarcely reported, with rare instances of radiation-induced retinopathy (44). The findings

piMTX									
0.43 (0.21,0.92)	oMTX+IVGCs								
0.38 (0.09,1.56)	0.87 (0.21,9.14)	Teprotumumab							
1.00 (0.79,6.06)	2.30 (0.31,16.69)	2.65 (0.65,10.77)	TA						
0.87 (0.31,11.46)	2.01 (0.73,16.53)	2.31 (0.70,13.62)	0.87 (0.32,8.40)	Rituximab					
0.84 (0.19,20.67)	1.94 (0.45,31.30)	**2.23 (1.01,14.91)**	0.84 (0.20,15.60)	0.97 (0.28,3.39)	OR				
0.90 (0.56,5.45)	2.07 (1.16,10.70)	2.38 (0.62,9.11)	0.90 (0.60,3.36)	1.03 (0.41,4.09)	1.07 (0.27,6.31)	OGC			
2.28 (0.24,31.40)	5.18 (0.87,44.45)	**5.49 (2.31,16.59)**	2.18 (0.31,22.61)	2.25 (0.56,11.15)	2.04 (0.55,4.79)	2.45 (0.54,16.52)	NT		
0.45 (0.25,0.82)	1.04 (0.60,4.81)	1.20 (0.32,7.53)	**0.45 (0.26,0.78)**	0.52 (0.21,1.29)	0.54 (0.14,3.14)	**0.50 (0.35,0.72)**	**0.27 (0.03,0.94)**	MMF+IVGCs	
0.60 (0.35,1.05)	1.38 (0.84,2.29)	1.59 (0.43,8.89)	**0.60 (0.36,0.99)**	0.69 (0.29,2.16)	0.71 (0.18,2.79)	**0.67 (0.50,0.89)**	0.36 (0.11,1.20)	**1.33 (1.07,1.65)**	IVGCs

Figure 2. Network meta-analysis was conducted to compare the overall response rate among different treatment modalities. Effect sizes represent summary the RRs (95% CIs) from the network meta-analysis between the column-defining intervention and row-defining intervention. RR >1 indicate that the intervention in the column is favored, while RR <1 indicate that the intervention in the row is favored.

Abbreviations: IVGCs, intravenous glucocorticoids; OR, orbital radiotherapy; OGC, oral glucocorticoid; MMF, mycophenolate mofetil; TA, triamcinolone acetonide; piMTX, periocular methotrexate; oMTX, oral methotrexate; RR, risk ratio.

IVGCs											
-1.92 (-5.72,1.97)	IVGCs+OR										
0.07 (-1.52,1.70)	2.00 (-2.22,6.14)	MMF+IVGCs									
-1.21 (-3.28,0.91)	0.70 (-2.51,3.93)	-1.29 (-3.89,1.39)	NT								
-0.81 (-1.06,-0.55)	0.83 (-3.33,4.86)	**-0.88 (-1.27,-0.49)**	0.12 (-2.45,2.57)	OGC							
-2.72 (-7.14,1.78)	-0.79 (-3.07,1.47)	-2.79 (-7.50,1.98)	-1.50 (-5.46,2.42)	-1.63 (-6.22,3.13)	OGC+OR						
-0.19 (-2.57,2.16)	1.72 (-2.85,6.18)	-0.27 (-3.15,2.55)	1.02 (-2.20,4.13)	0.89 (-1.83,3.64)	2.53 (-2.61,7.51)	oMTX+IVGCs					
-1.63 (-2.90,-0.36)	0.35 (-1.92,2.63)	**-2.29 (-4.17,-0.41)**	-0.34 (-2.64,1.95)	-0.47 (-3.82,3.00)	1.15 (-2.04,4.38)	-1.37 (-5.23,2.57)	OR				
-0.64 (-4.31,2.98)	1.28 (-4.10,6.49)	-0.71 (-4.71,3.24)	0.58 (-3.71,4.73)	0.45 (-2.93,3.82)	2.08 (-3.73,7.74)	-0.44 (-4.81,3.89)	0.92 (-3.92,5.66)	piMTX			
0.21 (-1.91,2.31)	2.14 (-1.78,5.96)	0.14 (-2.53,2.75)	1.44 (-0.76,3.53)	1.31 (-1.18,3.82)	2.94 (-1.59,7.36)	0.41 (-2.76,3.59)	1.78 (-1.39,4.87)	0.86 (-3.35,5.07)	Rituximab		
-0.84 (-3.60,1.88)	1.08 (-3.73,5.75)	-0.91 (-4.13,2.23)	0.38 (-3.16,3.76)	0.25 (-2.12,2.64)	1.87 (-3.44,7.03)	-0.64 (-4.28,2.96)	0.72 (-3.48,4.81)	-0.19 (-2.60,2.19)	-1.06 (-4.52,2.37)	TA	
0.35 (-2.69,3.46)	2.28 (-1.65,6.22)	0.27 (-3.17,3.75)	1.57 (-0.68,3.81)	1.44 (-1.86,4.86)	3.07 (-1.47,7.61)	0.55 (-3.30,4.48)	**1.93 (1.36,2.50)**	0.98 (-3.71,5.82)	0.13 (-2.91,3.29)	1.19 (-2.86,5.38)	Teprotumumab

Figure 3. Network meta-analysis was conducted to compare the change of CAS outcomes among different treatment modalities. Effect sizes represent summary the MDs (95% CIs) from the network meta-analysis between the column-defining intervention and row-defining intervention. MD <0 indicate that the intervention in the column is favored, while MD >0 indicate that the intervention in the row is favored.

Abbreviations: IVGCs, intravenous glucocorticoids; OR, orbital radiotherapy; OGC, oral glucocorticoid; MMF, mycophenolate mofetil; TA, triamcinolone acetonide; piMTX, periocular methotrexate; oMTX, oral methotrexate; MDs, mean differences.

highlighted safety concerns, with occurrences of adverse events being significantly more likely with OGCs (RR 4.7, 95% CI 1.1 to 24) than with placebo. These results are represented in the forest plot shown in Fig. 6. While the majority of adverse events were of mild to moderate severity, encompassing gastrointestinal issues and weight gain, there were reports of more serious adverse events. Specifically, 2 cases of intercostal neuralgia and optic neuropathy following teprotumumab treatment were documented in Kahaly et al (45), and 2 cases of pneumothorax (potentially unrelated to the medication) and infusion reactions post-teprotumumab were reported in Douglas (29). According to the study statistics, the adverse effects of teprotumumab mainly involve the muscle and bone (46). Additionally, discontinuation of oral glucocorticoids due to severe adverse reactions (possibly associated with higher doses) was noted in 4 cases in Macchia's 2001 study.

Comprehensive Network Meta-analysis

To comprehensively compare the efficacy and safety of these treatments, we utilized clustered ordinal plots based on the area under the SUCRA curve for overall efficacy and adverse events. This plot indicated that oMTX + IVGCs, MMF + IVGCs and teprotumumab provided an effective and safe treatment for GO, with an acceptable risk level for adverse effects (Fig. 7).

IVGCs											
0.10 (-1.31,1.53)	IVGCs+OR										
1.40 (0.20,2.63)	1.29 (-0.56,3.18)	MMF+IVGCs									
-0.24 (-2.96,2.42)	-0.33 (-3.50,2.63)	-1.64 (-4.61,1.25)	piMTX								
-0.48 (-1.91,0.94)	-0.60 (-2.57,1.51)	-1.90 (-3.79,0.07)	-0.26 (-3.31,2.85)	oMTX+IVGCs							
-1.02 (-3.08,0.96)	-1.13 (-2.79,0.46)	**-2.42 (-4.83,-0.11)**	-0.79 (-4.05,2.71)	-0.53 (-3.05,1.88)	NT						
-0.65 (-1.22,-0.11)	-0.74 (-2.27, 0.72)	-2.05 (-3.41,-0.75)	-0.41 (-3.02,2.25)	-0.16 (-1.73,1.38)	0.36 (-1.69,2.51)	OGC					
-0.21 (-1.99,1.65)	-0.29 (-1.45,0.82)	-1.59 (-3.80,0.64)	0.04 (-3.11,3.44)	0.29 (-2.08,2.61)	0.84 (-1.08,2.83)	0.45 (-1.40,2.39)	OGC+OR				
-0.91 (-2.61,0.82)	**-1.01 (-1.94,-0.09)**	**-2.33 (-4.47,-0.18)**	-0.69 (-3.82,2.65)	-0.42 (-2.67,1.75)	0.12 (-1.17,1.42)	-0.27 (-2.03,1.55)	-0.73 (-2.28,0.83)	OR			
-0.93 (-3.37,1.52)	-1.04 (-3.16,1.15)	-2.32 (-5.10,0.44)	-0.69 (-4.21,3.11)	-0.45 (-3.28,2.44)	0.11 (-1.25,1.50)	-0.28 (-2.80,2.28)	-0.73 (-3.09,1.71)	0.01 (-1.91,1.91)	Rituximab		
-0.71 (-2.25,0.81)	-0.79 (-2.92,1.20)	**-2.11 (-4.10,-0.21)**	-0.47 (-2.68,1.84)	-0.21 (-2.30,1.89)	0.33 (-2.22,2.86)	-0.05 (-1.49,1.36)	-0.50 (-2.97,1.78)	0.21 (-2.14,2.44)	0.22 (-2.72,3.13)	TA	
1.27 (-0.82,3.29)	1.15 (-0.54,2.81)	-0.12 (-2.56,2.21)	1.50 (-1.79,5.04)	1.75 (-0.79,4.20)	**2.29 (1.86,2.73)**	**1.93 (0.07,3.80)**	1.44 (-0.58,3.41)	**2.17 (0.80, 3.51)**	**2.18 (0.73,3.61)**	1.96 (-0.60, 4.55)	Teprotumumab

Figure 4. Network meta-analysis was conducted to compare the change of proptosis outcomes among different treatment modalities. MD <0 indicate that the intervention in the column is favored, while MD >0 indicate that the intervention in the row is favored.

Abbreviations: IVGCs, intravenous glucocorticoids; OR, orbital radiotherapy; OGC, oral glucocorticoid; MMF, mycophenolate mofetil; TA, triamcinolone acetonide; MTX, methotrexate; piMTX, periocular methotrexate; oMTX, oral methotrexate; MDs, mean differences.

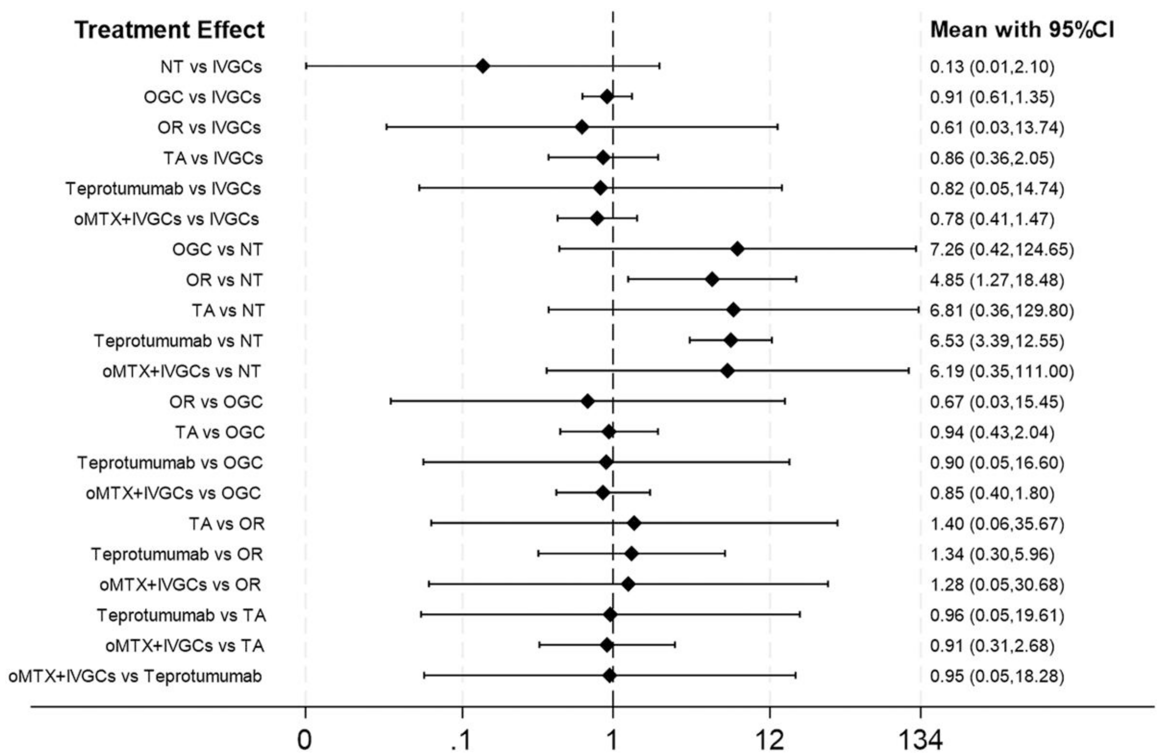

Figure 5. Forest plot of network effect sizes among different treatments for diplopia measured as percentages. If RR >1, the former option was more effective in comparison; if the RR <1, the latter option was more effective in comparison.

Abbreviations: IVGCs, intravenous glucocorticoids; OR, orbital radiotherapy; OGC, oral glucocorticoid; MMF, mycophenolate mofetil; TA, triamcinolone acetonide; piMTX, periocular methotrexate; oMTX, oral methotrexate; RR, risk ratio.

Quality of Trials and Risk of Bias

Upon quality assessment of the 19 studies, all were found to have an indeterminate risk of bias. Among these, 10 studies (53%) demonstrated a low risk of bias in random sequence generation, 10 studies in allocation concealment, 10 studies in double-blinding, and 15 studies (79%) in handling of missing outcome data. A detailed bias assessment is provided elsewhere (Appendix S4 (16)).

To evaluate publication bias, funnel plots for the main outcomes were created, and the Egger test and AS-Thompson test were conducted, revealing no significant asymmetry and no indication of bias towards studies with smaller sample sizes (Appendix S10 (16).

Evaluation of Consistency

Global inconsistency tests for the 5 main outcomes showed no significant discrepancies (Appendix S5 (16)). When using the Markov Chain Monte Carlo method to identify global consistency, we discovered better convergence and focused densities in the convergence diagnostics. Local node inconsistency tests for overall effectiveness and proptosis also indicated no evidence of inconsistency. For the heterogeneity analysis of overall response rate, we found no heterogeneity in most outcomes but moderate heterogeneity in others; for CAS, we found no heterogeneity in some outcomes but moderate heterogeneity in others. for proptosis, we found no heterogeneity (Appendix S6 (16)). Regression analysis was conducted to explore the sources of this heterogeneity, and no significant effect on the primary outcomes was found (Appendix S11 (16)).

According to our analysis, the heterogeneity in the CAS results could be attributed to the significant differences between the Salvi and Roy studies on the effectiveness of rituximab in decreasing CAS. These investigations did not, however, reveal any appreciable difference in the baseline parameters or the specifics of the intervention. Furthermore, it has been shown that when fewer studies are included in a meta-analysis, I^2 values and their confidence intervals are also greater (24).

Sensitivity analyses were carried out by excluding 1 or 2 studies that produced heterogeneity, we found some difference in MD with 95% CI and SUCRA rankings on CAS. Therefore, we reduced the confidence level of CAS by 1 rank, and careful consideration is needed for CAS rankings (Appendix S12 (16)).

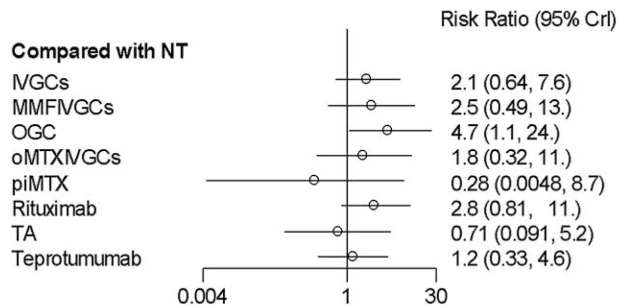

Figure 6. A placebo comparing the probability of adverse reactions to different drugs was used as effect size with risk ratio and 95% CI. If RR >1, the drug was more likely to cause an adverse reaction than placebo; if RR <1, the drug was safer than placebo.

Abbreviations: IVGCs, intravenous glucocorticoids; OGC, oral glucocorticoid; MMF, mycophenolate mofetil; TA, triamcinolone acetonide; piMTX, periocular methotrexate; oMTX, oral methotrexate; RR, risk ratio.

Discussion

Comprehensive Analysis

IVGCs, while effective for patients with GO, the emergence of new immunosuppressive drugs and combinations has demonstrated superior effectiveness, suggesting a shift towards these options for sustained remission.

Teprotumumab (ID:AB_2892149) shows a remarkable effectiveness in all 4 included RCTs. Yet, a notable proportion of patients treated with teprotumumab may experience relapses (47), necessitating further interventions and considering the high costs involved (48).

Mycophenolate salts, as inhibitors of immune cell and antibody proliferation (49), exhibit a high safety profile for treating active GO. When combined with IVGCs, mycophenolate can also achieve good therapeutic effects.

Rituximab (ID:AB_915401), targeting CD20-positive B cells, showed mixed results in its effectiveness (50). The 2 studies on rituximab included in this paper presented divergent results, with 1 trial showing no benefit compared to placebo (32) and another trial compared to IVGCs showing a significant effect in reducing CAS scores but no significant efficacy in improving ocular prominence (33).

Orbital radiotherapy is believed to reduce inflammation around the eye due to the radiosensitivity of lymphocytes that infiltrate the orbit (51). The findings of the research included in this article indicate that there was no statistically significant difference in improving GO activity and proptosis when comparing the effects of sham radiotherapy.

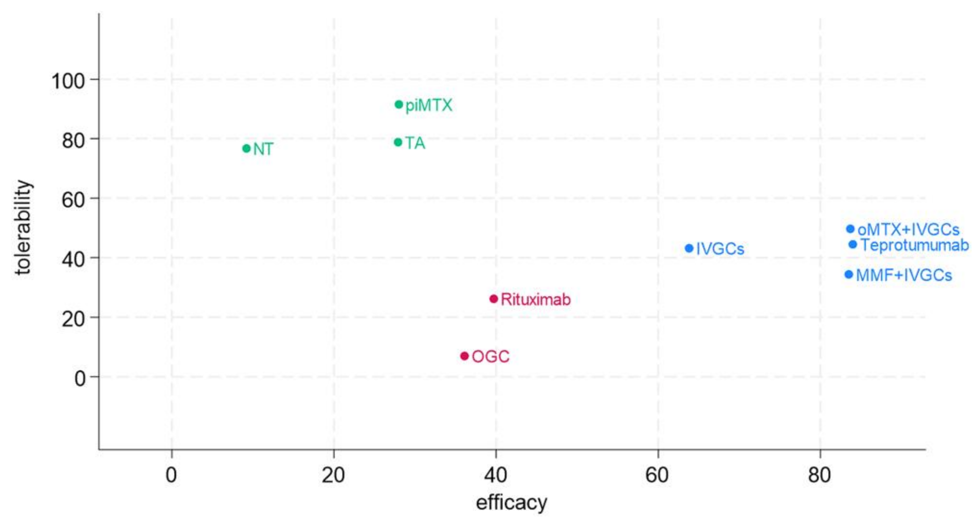

Figure 7. Comprehensive efficacy among different drugs. Each point is positioned according to the 2 SUCRA values for overall effectiveness and adverse events for each intervention. A point positioned further to the right implies a better ranking for efficacy, and a point positioned further to the top implies a better ranking for safety. Interventions located in the upper right corner had higher SUCRA values for both effectiveness and safety variables.

Abbreviations: IVGCs, intravenous glucocorticoids, OGC, oral glucocorticoid, MMF, mycophenolate mofetil; TA, triamcinolone acetonide; piMTX, periocular methotrexate; oMTX, oral methotrexate.

Nevertheless, there was a greater effectiveness in improving diplopia, and some studies have demonstrated that radiotherapy has a notable impact on the amelioration of ocular dyskinesia (52).

When methotrexate and triamcinolone acetonide are administered via the periocular route, both drugs are safe and effective. both drugs are effective and have a lower incidence of side effects (35, 53), particularly effective in reducing activity and soft tissue swelling. Moreover, significant efficacy was observed when oral methotrexate combined with IVGCs.

Strengths and Limitations of This Study

In contrast to the 2 previous reticulated meta-analyses (12, 54), which focused on the clinical effectiveness rate, the main objective of the 33 studies analyzed in Zhou et al (12) was to determine the clinical remission rate as defined by the authors. The study concluded that the most effective treatment was a combination of radiotherapy and glucocorticoids, followed by mycophenolate mofetil. The most recent RCTs and further studies on the novel medicine teprotumumab have been incorporated into the paper. Apart from comparing the overall effectiveness, we also provide support and help for the clinical treatment of patients with thyroid eye disease with different symptoms or different severity in presentation and also provides data support for the efficacy and safety of new drugs.

However, our study also has some limitations. Variability in population characteristics, follow-up durations, and baseline conditions across the included trials may have introduced imprecision in effect estimates. Furthermore, the subjective nature of CAS scores as an activity assessment tool highlights the need for more objective measures in future research. The exploration of novel indicators, such as ^{99}Tcm-DTPA SPECT/CT orbital imaging, could enhance the precision of activity evaluations and treatment response predictions.

Conclusion

In conclusion, this network meta-analysis provides a rigorous and comprehensive comparison of nonsurgical treatments for active GO, highlighting teprotumumab's prominence in overall efficacy. However, the potential for higher costs necessitates careful consideration and monitoring. The favorable balance between efficacy and safety observed mycophenolate or methotrexate combined with IVGCs, suggests these as viable alternatives, especially in cases where teprotumumab's use may be limited by safety concerns or cost considerations. The findings advocate for a personalized treatment approach, MMF + IVGCs is an acceptable choice for patients with red, swollen or photophobic eyes, MMF + IVGCs and teprotumumab are beneficial for obvious eye protrusion; teprotumumab and orbital radiotherapy are beneficial for blurred vision and diplopia. Future research should focus on new drugs with both efficacy and safety, as well as more clinical studies on existing treatment strategies with uncertain efficacy to further refine treatments for GO.

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Disclosures

The authors have nothing to disclose. Original data generated and analyzed during this study are included in this article or supplement material.

Data Availability

Some or all datasets generated during and/or analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request.

Meta-analysis registration number

PROSPERO (CRD42024548030).

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